



DPUDataFile

Recording of DPU output in text format.

1. Launchdata:

VAR_SAVEDATA

Flag defining if data should be recorded.

VAR_DATAFILENAME

Filename.

VAR_USERDATA[XX]

User data (not more than 10) that is included in the rec-orded data.

Remark: The names for launch data can be specified by *LD_UserDataToken[XX]*.

2. Inputs:

(generated)

The Record parameter describes all variables that will be recorded (see below). The appropriate input variables are created and connected to those variables automatically.

3. Dynamic parameters:

SaveData

Defines whether data is recorded at all. Default: 1 (data is recorded). Value zero defines no recording. Normally the launch data *VAR_SAVEDATA* is used instead (see below). However, the *SaveData* parameter can be used to temporarily disable the data recording during the simulation.

EventsOnly

Defines when data is recorded. The default value (0) records data continuously whenever *SaveData*=1. If *EventsOnly* is set to 1, only one line of data is recorded when-ever the *SaveData* value changes from 0 to 1.

4. Static parameters:

Header

= 1: A header is included in the recorded file.

= 0: No header is included in the recorded file.

HeaderType

Header format:

= 0: For each column <number> a line of the following type is exported: „column<number> =name of column “

= 1: A line containing all the names of columns is exported. The names are separated by a „Separator“ character (see below).

Path

Output path of recorded file. If no absolute path is provided the path starts in the installation path of SILAB.

FileName

The filename of the recorded file.

Extension

File extension. Default: „asc“

Separator

Character for separating columns.

LineNumber

Column number that contains the line number in the recorded file.

Default = 0. If the column number is within the user data columns or



SILAB DRIVING SIMULATION

version Version 7.1 documentation

	between measurement time and time error the line number is written after these outputs.
<i>LaunchDataPosition</i>	Column of user data VAR_USERDATA[XX] that can be set by the launch data dialog: 0 = in the starting columns (default), 1 = behind the measurement time column 2 = behind all other columns
<i>WriteConfig</i>	= 1: Writing the expanded configuration file additionally to the recording file. (Default). = 0: The expanded configuration file is not written in the output directory.
<i>OverwriteExistingFile</i>	Flag for overwriting an existing file. Default = false: Each file gets a unique name.
<i>Subsample</i>	If data should be recorded not every simulation step, "Subsample" defines the number of steps for writing a single data line. Default = 1 which means that data is recorded every simulation step.
<i>DiskFree_Warning_MB</i>	Minimum recommended free disk space (MB) before starting a simulation. If less space is free, a warning is issued. Default: 1000
<i>DiskFree_Error_MB</i>	Minimum required free disk space (MB) before starting a simulation. If less space is free, the simulation cannot be started, because the space will probably run out. Default: 10
<i>MD5</i>	Boolean flag for calculating a MD5-Hash. MD5 = 1: A file with the same name as the data file is created with the extension "MD5Extension". This file can be used to check if the recorded data file is still in original state. MD5 = 0 (default): No MD5-Hash is calculated.
<i>MD5Extension</i>	Extension for the MD5-Hash file. (Default = „md5“).
<i>MD5BlocksPerStep</i>	The hash file is created when pressing the stop button in SILAB. The computation is divided into multiple time steps. In every time step a number of data blocks is processed for calculating the MD5-Hash. Parameter „MD5BlocksPerStep“ defines this number of blocks. This parameter has to be chosen carefully in order to avoid time delays. (Default = 10000);
<i>LD_UserDataToken[01...10]</i>	Name of launch data for user specific data (not more than 10). Default: VAR_USERDATA[XX].
<i>UserDataAlias[01...10]</i>	Name of user data column in ASCII-file. (Default: UserXX)

5. Functional principle:

DPUDataFile is used for data recording in text format. The variables that are recorded can be specified in the "Record" parameter like this:

```
DPUDataFile DataFile {  
    Record = {  
        SourceDPUName.VarName,  
        SourceDPUName.AnotherVarName,  
        AnotherSourceDPUName.Var3 ("Alias")  
    }
```



```
};  
};
```

The “Record” parameter contains a comma-separated list of variables of any DPU.

For every variable an alias can be specified (in brackets) which is used as column name for the appropriate variable. If no alias is specified, the full name of the variable is used as name for the column.

In the example above, the columns in the recorded file are denoted by „SourceD-PUName.VarName“, „SourceDPUName.AnotherVarName“ and “Alias”.

In addition to all the variables that are specified for recording in the “Record” parameter, the measuring time as well as the measuring time error are recorded in the first columns.

The measuring time is the time since start of the simulation in milliseconds.

The measuring time error is the difference between the point in time that was planned for measuring and the point in time at which the variable was actually measured.

The filename of the output file is defined as follows:

Path\FileName.sequentialnumber .Extension

The sequential number is always increased if a file with the same name already exists. The current number is saved in an additional file with the extension .XXX.num.

WriteConfig = true also saves the expanded configuration (i.e. without include – statements) in an additional file. This file has the same name as the data file and the extension cfg.

6. Example:

A header can be added to the recorded data including all column names. Column one (measurement time in milliseconds) and column two (time error in milliseconds) are always added to the recorded variables. All other column names are specified by their alias or their variable name.

In order to create the following recording of six variables:

Example: Data recording of 6 variables:

```
Spalte1=measurement time  
Spalte2=time error  
Spalte3=steering angle  
Spalte4=brake pedal  
Spalte5=accelerator pedal  
Spalte6=speed[m/s]  
Spalte7=Y_acceleration[m/s*s]  
Spalte8=vYaw[rad/s]  
0 0 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000  
10 0 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000  
20 0 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000  
30 0 0.000000 0.000000 0.000000 0.000486 0.000000 0.000000
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
40 0 0.000000 0.000000 0.000000 0.001310 0.000000 0.000000
50 0 0.000000 0.000000 0.000000 0.002428 0.000000 0.000000
60 0 0.000000 0.000000 0.000000 0.003830 0.000000 0.000000
70 0 0.000000 0.000000 0.000000 0.005507 0.000000 0.000000
80 0 0.000000 0.000000 0.000000 0.007452 0.000000 0.000000
90 0 0.000000 0.000000 0.000000 0.009656 0.000000 0.000000
100 0 0.000000 0.000000 0.000000 0.012114 0.000000 0.000000
110 0 0.000000 0.000000 0.000000 0.014818 0.000000 0.000000
120 0 0.000000 0.000000 0.000000 0.017762 0.000000 0.000000
130 0 0.000000 0.000000 0.000000 0.020941 0.000000 0.000000
140 0 0.000000 0.000000 0.000000 0.024349 0.000000 0.000000
150 0 0.000000 0.000000 0.000000 0.027981 0.000000 0.000000
160 0 0.000000 0.000000 0.000000 0.031833 0.000000 0.000000
170 0 0.000000 0.000000 0.000000 0.035900 0.000000 0.000000
180 0 0.000000 0.000000 0.000000 0.040177 0.000000 0.000000
190 0 0.000000 0.000000 0.000000 0.044661 0.000000 0.000000
...
```

The following configuration can be used to create this recording:

```
DPUDataFile DataFile
{
    Computer = {MAIN};
    Index = 2000; # after all other DPUs
    Separator = " ";
    Record = {
        VDyn.SteeringWheel ("steering angle"),
        VDyn.BrakePedal ("brake pedal"),
        VDyn.AcceleratorPedal ("accelerator pedal"),
        VDyn.v ("speed[m/s]"),
        VDyn.ay ("Y_acceleration[m/s*s]"),
        VDyn.vYaw ("vYaw[rad/s]")
    };
};
```

7. Launch data configuration:

In order to avoid overwriting existing data recordings, it is often desired to ask for a unique file name each time the simulation is started. Launch data can be used to ask for this file name. The launch data definition has to be added to the SILAB System section of the configuration file.

Example of a launch data configuration

```
SILAB System
{
    LaunchData1 = ("Data_recording.",
        VAR_SAVEDATA, LD_BOOL_DEFAULT, "1");
}
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
LaunchData2 = ("Filename",  
               VAR_DATAFILENAME, LD_EDIT_DEFAULT, "Data");  
};
```

The launch data will be queried prior to the launch step.

If it is desired to include personal data of the subject in the recording, this data can also be entered using launch data:

Example: User data configuration

```
SILAB System  
{  
    LaunchData1 = ("Driver", VAR_USERDATA01, LD_EDIT_DEFAULT, "-9999");  
    LaunchData2 = ("Condition", VAR_USERDATA02, LD_EDIT_DEFAULT, "-9999");  
    LaunchData3 = ("Free1", VAR_USERDATA03, LD_EDIT_DEFAULT, "-9999");  
    LaunchData4 = ("Free2", VAR_USERDATA04, LD_EDIT_DEFAULT, "-9999");  
};  
  
...  
DPUDataFile DPUDataFile  
{  
    Computer = {LOCALHOST};  
    Index = 100;  
    LineNumber = 2;  
    HeaderType = 0;  
    Record = {DPUAMP.Out};  
};
```

Next you can see the recorded data file of the example above:

```
Column1=User01  
Column2=User02  
Column3=User03  
Column4=User04  
Column5=Row  
Column6=MeasurementTime  
Column7=MeasurementTimeError  
Column8=DPUAMP.Out  
-9999,-9999,-9999,-9999,1,0,0,0.000000  
-9999,-9999,-9999,-9999,2,10,0,0.000000  
-9999,-9999,-9999,-9999,3,20,0,0.000000  
-9999,-9999,-9999,-9999,4,30,0,0.000000  
-9999,-9999,-9999,-9999,5,40,0,0.000000  
-9999,-9999,-9999,-9999,6,50,0,0.000000  
-9999,-9999,-9999,-9999,7,60,0,0.000000
```

Modular recording



If the configuration becomes more complex, and trials are run with different DPU pools, then not all relevant variables might be available in each of the DPU pools.

An example is a study that compares between a configuration without a driver assistance system (DPU pool “Base”) and an extended configuration with the system (DPU pool “Assisted”).

For trials run with the “Base” DPU pool, you may want to define a recording that contains the driver’s inputs, e.g.:

```
DPUPool Base
{
    ...

    DPUDataFile DataFile
    {
        Computer = {LOCALHOST};
        Index = 100;
        LineNumber = 2;
        HeaderType = 0;
        Record = {
            Mockup.AcceleratorPedal("AcceleratorPedal"),
            Mockup.BrakePedal("BrakePedal"),
            Mockup.SteeringWheel("SteeringWheel"),
            VDyn.v_kmh("v_kmh")
        };
    };
};
```

For trials run with the “Assisted” pool, you may want to additionally record outputs of the assistance system. This is possible by extending the DataFile’s variable set using the += operator as follows:

```
DPUPool Assisted : Base
{
    ...

    DataFile.Record += {
        System.State("SystemState"),
        System.SetSpeed("SetSpeed")
    };
};
```